Modeling Strange Attractors Physics 77 Project Proposal

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A strange attractor is a type of attractor whose trajectory is greatly influenced by its initial state, doesn't repeat motion, and never settles to an equilibrium point. As a result of their sensitive dependence on their initial conditions, one small deviation in the starting configuration could have a tremendous impact on the resulting system while still retaining the general shape of the attractor. One notable example of a strange attractor is the Lorenz attractor, which is a simplified model of atmospheric conditions and resembles the shape of a butterfly. Its creator, Edward N. Lorenz, accidentally found that small changes to the initial conditions due to computer rounding led to vastly different outcomes.

Other examples of strange attractors we'll model are the Thomas, Langford, and Rössler attractors. For each, we want to show different phase spaces for small differences in the initial conditions and how they uniquely evolve over time. We will use Python to model these attractors through matplotlib, the utilization of NumPy, and a PyQt5 GUI. The values are plotted by inserting into lists the x, y, and z values after each change dx, dy, and dz respectively is applied for a small time step a certain N times according to the following differential equations:

Lorenz System:

Default parameters: $\sigma = 10,\ \rho = 28,\ \beta = \frac{8}{3}$

$$\frac{dx}{dt} = \sigma(y - x)$$

$$\frac{dy}{dt} = -xz + px - y$$

Langford:

Default parameters: $a = 0.95, b = 0.7, c = 0.6, d = 3.5, e = 0.25, f = 0.1$

$$\frac{dx}{dt} = (z - b)x - dy$$

$$\frac{dy}{dt} = dx + (z - b)y$$

$$\frac{dz}{dt} = c + az - \frac{z^3}{3} - (x^2 + y^2)(1 + ez) + fzx^3$$

Thomas:

Default parameter: $b = 0.208186$

$$\frac{dx}{dt} = \sin y - bx$$

$$\frac{dy}{dt} = \sin z - by$$

$$\frac{dz}{dt} = \sin x - bz$$

Rössler System:

Default parameters: $a = 0.2,\ b = 0.2,\ c = 5.7$

$$\frac{dx}{dt} = -(y + z)$$

$$\frac{dy}{dt} = x + ay$$

$$\frac{dz}{dt} = b + z(x - c)$$

Through our models we intend to show how a system, though deterministic, can still be unpredictable.

Campuzano, Juan Carlos Ponce. 2018. "Strange Attractors." Interactive Resource. Strange Attractors, July 1. https://jcponce.github.io/.